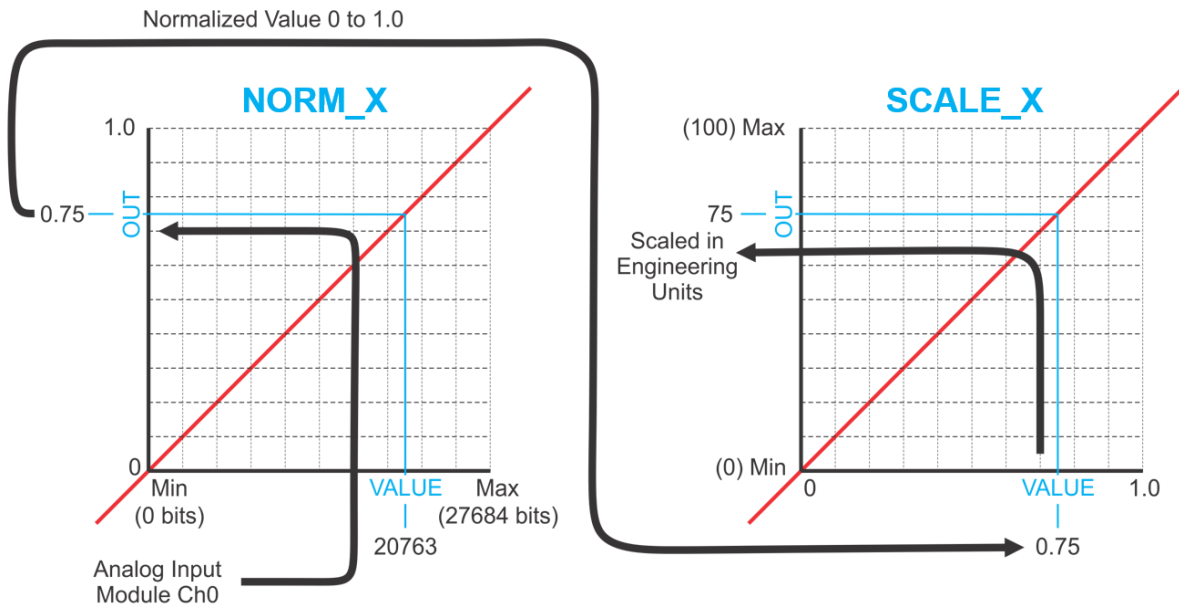


SIEMENS ANALOG

The NORMALIZE and SCALE instructions are used for converting the bit value to Engineering Unit for analog input and Engineering Unit to bit value for analog output.



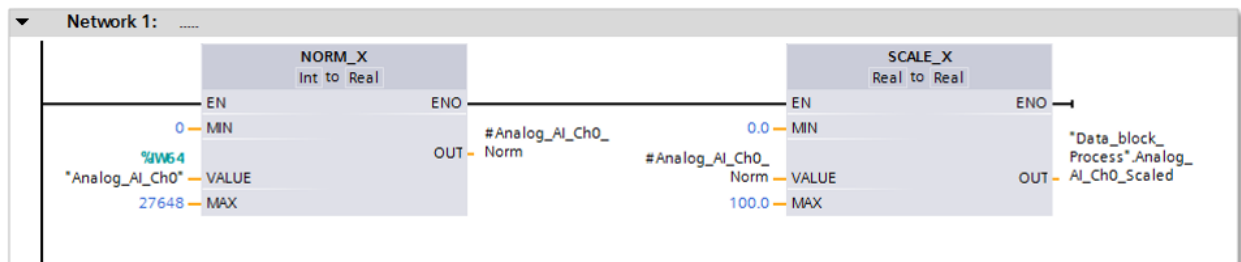
The normalize creates a "percentage value" that is applied to the scaled value.

LADDER LOGIC EXAMPLE

Analog Input

Bit Value: 0 to 27648

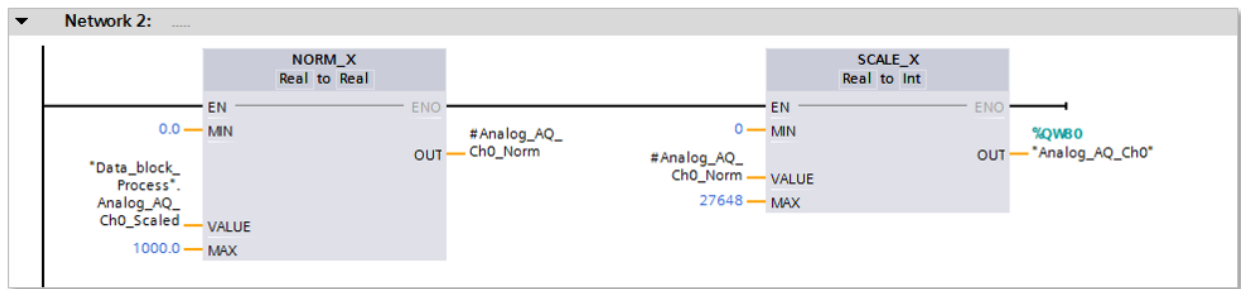
Engineering Units: 0 to 100



Analog Output

Engineering Units: 0 to 100

Bit Value: 0 to 27648

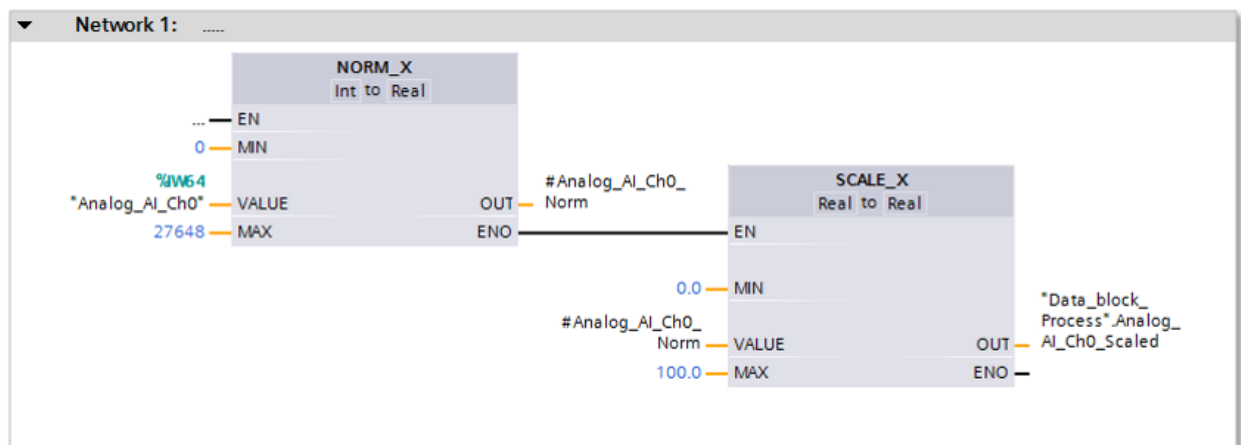


FUNCTION BLOCK EXAMPLE

Analog Input

Bit Value: 0 to 27648

Engineering Units: 0 to 100



STRUCTURED TEXT EXAMPLE

Analog Input

Bit Value: 0 to 27648

Engineering Units: 0 to 100

```
1 #Analog_AI_Ch0_Norm := NORM_X(MIN := 0,  
2     VALUE := "Analog_AI_Ch0",  
3     MAX := 27684);  
4 "Data_block_Process".Analog_AI_Ch0_Scaled := SCALE_X(MIN:=0,  
5     VALUE:=#Analog_AI_Ch0_Norm,  
6     MAX:=100);
```